<u>Harmonic oscillator</u> $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2$

$$\Rightarrow S = \int_0^T dt \left(\frac{1}{2} m \dot{x}^2(t) - \frac{1}{2} m\omega^2 x^2(t) \right) = \int dt\, L$$

$$D(x', T; x, 0) = \int D x(t)\, e^{\frac{i}{\hbar} S}$$

$$L = \frac{1}{2} m \dot{x}^2 - \frac{1}{2} m\omega^2 x^2$$

$$\frac{\partial L}{\partial x} = \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \Rightarrow m\ddot{x} = -m\omega^2 x \Rightarrow \ddot{x}(t) = -\omega^2 x(t)$$

So let us parameterize $x(t) = x_{cl}(t) + \delta x(t)$

where $x_{cl}(t)$ satisfies $\ddot{x}_{cl} = -\omega^2 x$ <u>and</u> BC $\begin{cases} x_{cl}(0) = x \\ x_{cl}(T) = x' \end{cases}$

$$\Rightarrow S(x(t)) = \int dt\, \frac{1}{2} m \left\{ (\dot{x}_{cl} + \delta\dot{x})^2 - \omega^2 (x_{cl} + \delta x)^2 \right\} =$$

$$= \int dt \left\{ \frac{m}{2} (\dot{x}_{cl}^2 - \omega^2 x_{cl}^2) + m(\dot{x}_{cl}\, \delta\dot{x} - \omega^2 x_{cl}\, \delta x) + \frac{m}{2} (\delta\dot{x}^2 - \omega^2 \delta x^2) \right\}$$

But $\int_0^T dt\, \dot{x}_{cl}\, \delta\dot{x} = \int_0^T dt \left\{ \frac{d}{dt} (\dot{x}_{cl}\, \delta x) - \ddot{x}_{cl}\, \delta x \right\} =$

$$= \underbrace{\dot{x}_{cl}\, \delta x \Big|_0^T}_{\substack{\| \\ 0 \text{ due to BC } \delta x(0) = \delta x(T) = 0.}} + \int dt \left(-\ddot{x}_{cl}(t)\, \delta x(t) \right)$$

$\Rightarrow$ the middle (linear in δx) term becomes

$$\int_0^T dt (-m) \underbrace{(\ddot{x}_{cl} + \omega^2 x_{cl})}_{\substack{\| \\ 0 \text{ by eqn of motion}}} \delta x = 0$$

$$\Rightarrow S(x = x_{cl} + \delta x) = \underbrace{\int_0^T dt \, \frac{m}{2}\left(\dot{x}_{cl}^2 - \omega^2 x_{cl}^2\right)}_{S[x_{cl}(t)]}$$

$\leftarrow$ classical action

$$+ \underbrace{\int_0^T dt \, \frac{m}{2}\left[\left(\delta\dot{x}(t)\right)^2 - \omega^2\left(\delta x(t)\right)\right]}_{S[\delta x(t)]}$$

$\leftarrow$ action of quantum fluctuations

Note that $\int D x(t) = \int D\,\delta x(t)$

$$\Rightarrow D(x', T; x, 0) = \underbrace{e^{\frac{i}{\hbar} S[x_{cl}]}}_{\text{task 1}} \underbrace{\int D\delta x(t)\, e^{\frac{i}{\hbar} S[\delta x]}}_{\text{task 2}}.$$

<u>task 1</u>

Clearly $x_{cl}(t) = A \sin(\omega t + \varphi)$, but we need to express A and φ in terms of initial and final points x, x':

$$\begin{cases} x_{cl}(0) = x = A \sin\varphi \\ x_{cl}(T) = x' = A \sin(\omega T + \varphi) \end{cases}$$

$$S_{cl} = \int_0^T dt \, \frac{m}{2}\left\{\left(A\omega \cos(\omega t + \varphi)\right)^2 - \omega^2\left(A \sin(\omega t + \varphi)\right)^2\right\}$$

$$= \frac{1}{2} m A^2 \omega^2 \int_0^T dt \, \cos(2\omega t + 2\varphi) =$$

$$= \frac{1}{4} m A^2 \omega \int_0^{2\omega T} d(2\omega t) \, \cos(2\omega t + 2\varphi) =$$

$$= \frac{1}{4} m\omega A^2 \left[\sin(2\omega T + 2\varphi) - \sin(2\varphi)\right].$$

$$\sin(2\varphi) = 2\sin\varphi\cos\varphi = 2\cos\varphi\,\frac{x}{A}$$

$$\sin(2\omega T+2\varphi) = 2\cos(\omega T+\varphi)\sin(\omega T+\varphi) = 2\cos(\omega T+\varphi)\,\frac{x'}{A}$$

$$\Rightarrow S_{cl} = \frac{1}{4}m\omega\cdot 2A\left[x'\underbrace{\cos(\omega T+\varphi)}_{\cos\omega T\cos\varphi-\sin\omega T\sin\varphi} - x\cos\varphi\right] =$$

$$= \frac{1}{2}m\omega\left\{x'\left(\cos\omega T\,(A\cos\varphi) - \sin\omega T\,\underbrace{(A\sin\varphi)}_{x}\right) - x\,(A\cos\varphi)\right\}$$

$\Rightarrow$ we need to express $A\cos\varphi$ in terms of x, x', T:

$$x' = A\sin(\omega T+\varphi) = A\left(\sin\omega T\cos\varphi + \cos\omega T\sin\varphi\right) =$$

$$= \sin\omega T\,(A\cos\varphi) + \cos\omega T\,x$$

$$\Rightarrow A\cos\varphi = \frac{x' - x\cos\omega T}{\sin\omega T}$$

So that

$$S_{cl} = \frac{1}{2}m\omega\left\{(x'\cos\omega T - x)\,\frac{x'-x\cos\omega T}{\sin\omega T} - x'x\sin\omega T\right\} =$$

$$= \frac{1}{2}m\omega\left\{\left((x')^2+x^2\right)\frac{\cos\omega T}{\sin\omega T} - x'x\underbrace{\left(\frac{\cos^2\omega T}{\sin\omega T} + \frac{1}{\sin\omega T} + \sin\omega T\right)}_{\frac{\cos^2\omega T+1+\sin^2\omega T}{\sin\omega T} = \frac{2}{\sin\omega T}}\right\}$$

$$S_{cl} = \frac{m\omega}{2}\left\{\left(x^2+x'^2\right)\cot\!an(\omega T) - 2xx'\,\frac{1}{\sin\omega T}\right\}$$

Check the limit: $S_{cl} \xrightarrow[\omega\to 0]{} \frac{m}{2}\left(\frac{x^2+x'^2}{T} - \frac{2xx'}{T}\right) =$

$= \frac{1}{2}m\,\frac{(x-x')^2}{T}$ is the classical action of a free particle

<u>task 2</u> Calculate $F = \int D\delta x(t)\, e^{\frac{i}{\hbar} S[\delta x]}$

Observe that δx obeys zero boundary conditions:

$$\delta x(0) = \delta x(T) = 0$$

For convenience of writing let $\delta x(t) \equiv y(t)$.

By def:

$$F = \lim_{N\to\infty} \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{N/2} \int_{-\infty}^{\infty} dy_1 \dots dy_j \dots dy_{N-1} \cdot$$

$$\cdot \exp\left\{ \underbrace{\frac{im}{2\hbar}\varepsilon \sum_{j=1}^{N}}_{\int dt'} \left[\underbrace{\left(\frac{y_j - y_{j-1}}{\varepsilon}\right)^2}_{(\delta\dot{x})^2} - \underbrace{\omega^2 y_j^2}_{\omega^2 \delta x^2} \right] \right\} =$$

$$= \lim_{N\to\infty} \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{N/2} \int \prod_j dy_j \; e^{-\frac{m}{2i\varepsilon\hbar} \sum_{j=1}^{N} \left[(y_j - y_{j-1})^2 - \varepsilon^2\omega^2 y_j^2\right]}$$

Now observe that we can re-write

$\sum_j [\quad]$ as $\left(\vec{z}^{\,T} A \vec{z}\right)$ where $\vec{z} = (y_1, y_2 \dots y_{N-1})^T$ is a column-vector

and A is $(N-1)\times(N-1)$ matrix

$$A_{N-1} = \begin{array}{c} \\ \\ \\ \\ \\ \\ \\ \end{array} \begin{array}{cc} \begin{pmatrix} 2-\varepsilon^2\omega^2 & -1 & 0 & \cdots & 0 \\ -1 & 2-\varepsilon^2\omega^2 & -1 & 0 \cdots & \vdots \\ 0 & -1 & 2-\varepsilon^2\omega^2 & \cdots & \vdots \\ \vdots & & & \ddots & \vdots \\ \vdots & & & 2-\varepsilon^2\omega^2 & -1 \\ 0 & \cdots & \cdots & -1 & 2-\varepsilon^2\omega^2 \end{pmatrix} & \begin{array}{c} 1 \\ 2 \\ 3 \\ \vdots \\ N-2 \\ N-1 \end{array} \end{array}$$

$\longleftarrow (N-1) \longrightarrow$ $\quad \downarrow \frac{T}{N}$

$$\Rightarrow \quad F = \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{N/2} \int d^{N-1}\vec{z}\; e^{-\frac{m}{2i\varepsilon\hbar}\vec{z}^T A \vec{z}} =$$

$$= \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{N/2} \frac{(2\pi)^{(N-1)/2}}{\left[\det\left(\frac{m}{i\varepsilon\hbar}A_{N-1}\right)\right]^{1/2}} = \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{N/2}\left(\frac{2\pi}{\frac{m}{i\varepsilon\hbar}}\right)^{\frac{N-1}{2}}(\det A)^{-1/2}$$

$$= \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{\frac{N}{2}}\left(\frac{2\pi i\varepsilon\hbar}{m}\right)^{\frac{N}{2}-\frac{1}{2}}(\det A_{N-1})^{-1/2} =$$

$$= \left(\frac{m}{2\pi i\hbar\varepsilon \det A_{N-1}}\right)^{1/2}$$

Using the recursion relation (see next page)

$$\det A_j = a_j \det A_{j-1} - c_{j-1} b_{j-1} \det A_{j-2}$$

$$(a_j = 2 - \varepsilon^2\omega^2, \quad c_{j-1} = b_{j-1} = -1)$$

$$\det A_j - 2\det A_{j-1} + \det A_{j-2} = -\varepsilon^2\omega^2 \det A_{j-1}$$

Introduce $g_j = \varepsilon \det A_j$

$$g_j - 2g_{j-1} + g_{j+2} = -\varepsilon^2\omega^2 g_{j-1}$$

In the limit $N \to \infty$ (which is what we look for)

$$g_j = g(j\varepsilon), \quad g_{j-1} = g(j\varepsilon - \varepsilon) = g(j\varepsilon) - \varepsilon g'(j\varepsilon) + \tfrac{1}{2}\varepsilon^2 g''(j\varepsilon)$$

$$g_{j-2} = g(j\varepsilon - 2\varepsilon) = g(j\varepsilon) - 2\varepsilon g'(j\varepsilon) + \tfrac{1}{2}(2\varepsilon)^2 g''$$

$$\Rightarrow \quad g - 2\left(g - \varepsilon g' + \tfrac{1}{2}\varepsilon^2 g''\right) + g - 2\varepsilon g' + 2\varepsilon^2 g'' = -\varepsilon^2\omega^2 g$$

$$\varepsilon^2 g'' = -\varepsilon^2\omega^2 g \quad \Rightarrow \quad g'' + \omega^2 g = 0$$

$$|A^{(n)}| = \begin{vmatrix} a_1 & b_1 & 0 & 0 & \\ c_1 & a_2 & b_2 & 0 & \\ 0 & c_2 & a_3 & b_3 & \\ & & \ddots & \ddots & \ddots \end{vmatrix}$$

tridiagonal matrix

$$|A^{(n)}| = \begin{vmatrix} \ddots & & & 0 \\ & A^{(n-1)} & & \\ & & \ddots & b_{n-1} \\ 0 & \cdots & c_{n-1} & a_n \end{vmatrix} = \begin{vmatrix} \ddots & & & & \\ & A^{(n-2)} & & & \\ & & \ddots & b_{n-2} & 0 \\ & & c_{n-2} & a_{n-1} & b_{n-1} \\ & & 0 & c_{n-1} & a_n \end{vmatrix} =$$

$(n \times n)$

$$= (-1)^{n-1+n} c_{n-1} \underbrace{\begin{vmatrix} A^{(n-2)} & \vdots & \\ & & 0 \\ \cdots 0 & c_{n-2} & b_{n-1} \end{vmatrix}}_{(n-1)\times(n-1)} + (-1)^{n+n} a_n \underbrace{\begin{vmatrix} A^{(n-2)} & & 0 \\ & & b_{n-1} \\ 0 & c_{n-2} & a_{n-1} \end{vmatrix}}_{(n-1)\times(n-1)}$$

$$= (-1)^{2n-1} c_{n-1} (-1)^{2(n-1)} b_{n-1} |A^{(n-2)}| + a_n |A^{(n-1)}|$$

$$\Rightarrow \det A^{(n)} = a_n \det A^{(n-1)} - c_{n-1} b_{n-1} \det A^{(n-2)} \quad (*)$$

$$\det A^{(2)} = \begin{vmatrix} a_1 & b_1 \\ c_1 & a_2 \end{vmatrix} = a_2 (a_1) - c_1 b_1$$

$$\Rightarrow \underbrace{\det A^{(1)} = a_1 \ , \ \det A^{(0)} = 1}_{\text{these are our boundary conditions for recursion relation } (*)}$$

Boundary conditions: $\det A_1 = 2-\varepsilon^2\omega^2$, $\det A_0 = 1$

$$\Rightarrow g_1 = \varepsilon \det A_1 \xrightarrow[\varepsilon\to 0]{} \varepsilon(2-\varepsilon^2\omega^2) = 0$$

$$\| $$

$$g(\varepsilon) \xrightarrow[\varepsilon\to 0]{} g(0)$$

But $\dot g(0) = \lim_{\varepsilon\to 0} \dfrac{\varepsilon \det A_1 - \varepsilon \det A_0}{\varepsilon} = \lim_{\varepsilon\to 0}(2-\varepsilon^2\omega^2-1) = 1$

$$\Rightarrow \quad g(0)=0 \ , \ \dot g(0) = 1$$

$$\Rightarrow g(t) = A\cos\omega t + B\sin\omega t \Rightarrow A=0$$

$$\dot g = -A\omega \cancel{\sin\omega t} + B\omega\cos\omega t = 1 \text{ for } t=0$$

$$\Rightarrow B = \frac{1}{\omega}$$

$$\Rightarrow g(t) = \frac{\sin\omega t}{\omega}$$

We need $\varepsilon \det A_{N-1} \xrightarrow[\substack{N\to\infty \\ \varepsilon\to 0}]{} g(T) = \dfrac{\sin\omega T}{\omega}$.

So that $F(T) = \left(\dfrac{m\omega}{2\pi i\hbar \sin(\omega T)}\right)^{1/2}$

Thus $D(x',T;x,0) = \sqrt{\dfrac{m\omega}{2\pi i\hbar \sin(\omega T)}}\; e^{-\frac{m\omega}{2i\hbar}\frac{(x^2+x'^2)\cos\omega T - 2xx'}{\sin\omega T}}$